Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2008-II-Q03] | Part A (Q1-Q5)

Answer: 729

1. Each cut reduces one dimension by one, so the remaining three dimensions sum to $37 - 10 = 27$. By AM–GM their product is at most 9^3.

2. The dimensions $9, 9, 9$ are attained by cuts $1, 4, 5$ in the respective directions. Maximum volume is 729.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2011-II-Q09] | Part B (Q6-Q10)

Answer: 559

1. Call the given two-term sum r and the target sum s. Expansion gives $r + s = (x_1 + x_4)(x_2 + x_5)(x_3 + x_6)$.

2. These three nonnegative factors sum to one. AM-GM gives $s \leq 1/27 - r \leq 19/540$.

3. The bound is attained by $(x_1, \ldots, x_6) = (1/3, 19/60, 1/3, 0, 1/60, 0)$. All opposite-pair sums are $1/3$ and $r = (1/3)^2(1/60) = 1/540$.

4. Thus the maximum is $19/540$, giving 559.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2016-I-Q09] | Part B (Q6-Q10)

Corrected result: $\dfrac{248(1600-496\sqrt{6})}{2561-992\sqrt{6}} \approx 728.365125470659$

Historical answer key: 744

The historical source key is 744. Its equality configuration does not satisfy B on segment QR. With all original segment conditions retained, the maximum is the exact value shown above, attained when B=R. This is an independently documented source discrepancy; no official acknowledgment of an erratum is claimed.

1. Put $A=(0,0), Q=(w,0), R=(w,h), S=(0,h), B=(w,b), C=(c,h)$ with $0 \le b \le h, 0 \le c \le w$. Let $\beta=\angle BAQ$, $\gamma=\angle CAS$, and $\alpha=\angle BAC$. The triangle lies inside a rectangle corner, so α is acute and $\cos\alpha = 2\sqrt{6}/5$. Also $\beta+\gamma = 90^\circ - \alpha$.

2. The side lengths give $w = 40\cos\beta$, $b = 40\sin\beta$, $c = 31\sin\gamma$, $h = 31\cos\gamma$. Consequently $K = wh = 620[\cos(\beta-\gamma)+1/5] \le 744$. However equality needs $\beta=\gamma$, where $\tan\beta = \sqrt{2/3}$. Then $b \le h$ would require $40\sqrt{2/3} \le 31$, which is false. Thus 744 is an unattainable upper bound for the stated segment problem.

3. Impose $b \le h$: using $\cos\gamma = \sin(\alpha+\beta)$ gives $\tan\beta \le 31/(200-62\sqrt{6})$. The other condition is automatic since $c = 31\cos(\alpha+\beta) \le 31\cos\alpha\cos\beta < 40\cos\beta = w$. The feasible interval is therefore $0 \le \beta \le \arctan[31/(200-62\sqrt{6})] \approx 32.78425^\circ$.

4. The unconstrained area increases until $\beta = (90^\circ-\alpha)/2 \approx 39.23152^\circ$, beyond that feasible interval. Hence the constrained maximum occurs at its right endpoint, where $b=h$ and $B=R$.

5. At this endpoint, $d = BC$ is horizontal and cosine law gives $d^2 = 2561 - 992\sqrt{6}$. The determinant and dot product of AB, AC give $hd = 40\cdot 31\sin\alpha = 248$ and $wd = 40^2 - 40\cdot 31\cos\alpha = 1600 - 496\sqrt{6}$.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2016-I-Q09] | Part B (Q6-Q10) (continued / 续)

6. Thus $K_{\max} = 248(1600 - 496\sqrt{6})/(2561 - 992\sqrt{6}) \approx 728.365125470659$. It is attained by $h = 248/d, w = (1600 - 496\sqrt{6})/d, C = (w - d, h), B = R$; these satisfy all original lengths and containment conditions. The value 744 remains recorded only as the historical source key.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2022-II-Q06] | Part B (Q6-Q10)

Answer: 841

1. Total positive mass and total negative magnitude are each $1/2$. Sixteen terms no larger than x_{16} imply $x_{16} \geq -1/32$; twenty-five terms no smaller than x_{76} imply $x_{76} \leq 1/50$.

2. Therefore the difference is at most $1/32 + 1/50 = 41/800$. Equality occurs with sixteen copies of $-1/32$, fifty-nine zeros, and twenty-five copies of $1/50$.

3. This ordered tuple satisfies both constraints, so the bound is the maximum and $m + n = 841$.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2005-I-Q11] | Part C (Q11-Q15)

Answer: 544

1. For a unit-radius semicircle whose diameter forms angle $\theta \in [0, \pi/2]$ with a square side, its two perpendicular projection widths are $1 + \cos\theta$ and $1 + \sin\theta$. Thus the least square side for that orientation is their maximum.

2. This maximum is minimized at $\theta = \pi/4$, with value $1 + 1/\sqrt{2}$. Scaling to side eight gives $d = 16/(1 + 1/\sqrt{2}) = 32 - 16\sqrt{2} = 32 - \sqrt{512}$.

3. Therefore $m + n = 544$.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2008-I-Q14] | Part C (Q11-Q15)

Answer: 432

1. Use circle center $(0,0)$, $A=(-9,0), B=(9,0)$. A tangent has equation $ux+vy=9$ with $u^2+v^2=1$ and $-1<u<0$ because its axial intercept is beyond A. The projection is $P=A+9(1+u)(u,v)$.

2. Expand $BP^2/81=(u^2+u-2)^2+(1+u)^2(1-u^2)=5-2u-3u^2$. Its maximum on $(-1,0)$ occurs at $u=-1/3$, value $16/3$.

3. Thus $m^2=81\cdot 16/3=432$.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2009-II-Q15] | Part C (Q11-Q15)

Answer: 014

1. Let intersections be $Q = AC \cap MN$, $P = BC \cap MN$ and $x = CM/CN > 0$. Pythagoras gives $BN = 4/5$, while $AM = AN$. Cyclic sine-area ratios yield $MP/PN = BM \cdot CM/(BN \cdot CN) = 3x/4$ and $MQ/QN = x$.

2. As $MN = 1$, $d = x/(x+1) - 3x/(3x+4) = 1/(3x + 4/x + 7)$. The full opposite semicircle makes x range over all positive values.

3. AM–GM gives $3x + 4/x \geq 4\sqrt{3}$, with equality at $x = 2/\sqrt{3}$, an attainable ratio. Hence maximum $d = 1/(7 + 4\sqrt{3}) = 7 - 4\sqrt{3}$, giving 14.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2016-II-Q15] | Part C (Q11-Q15)

Answer: 863

1. The square of $\sum x_i = 1$ converts the second condition into $\sum_i x_i^2/(1-a_i) = 1/215$. All denominators are positive.

2. The geometric sum gives $\sum a_i = 1$, so $\sum(1-a_i) = 215$. Weighted Cauchy therefore gives $1 = (\sum x_i)^2 \le [\sum x_i^2/(1-a_i)]\sum(1-a_i) = 1$.

3. Equality forces $x_i/(1-a_i)$ constant, giving the unique positive tuple $x_i = (1-a_i)/215$. It satisfies both original conditions. Thus $x_2 = 3/860$ is its only, hence maximum, value, and $m+n = 863$.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2024-I-Q15] | Part C (Q11-Q15)

Answer: 721

1. For dimensions $a, b, c > 0$, $ab + bc + ca = 27, abc = 23$. The required squared radius is one fourth the maximum space-diagonal square, namely $[(a + b + c)^2 - 54]/4$.

2. A maximum exists: each dimension is at least $23/27$, while $27 \geq a(b + c) \geq 2\sqrt{23a}$ gives $a \leq 729/92$, and similarly for the others. The constrained positive dimensions therefore form a compact set.

3. At an extremum two dimensions coincide: otherwise the cubic with roots a, b, c could have its sum perturbed while maintaining three positive roots. Setting $b = c = t$ gives $t^3 - 27t + 46 = (t - 2)(t^2 + 2t - 23) = 0$. The positive candidates are $t = 2$ and $t = 2\sqrt{6} - 1$.

4. For $t = 2$, $(a, b, c) = (23/4, 2, 2)$ and the sum is $39/4$. For the second candidate the sum is $(-21 + 96\sqrt{6})/23 < 39/4$ (use $\sqrt{6} < 5/2$, giving $s < 219/23 < 39/4$). Thus $r^2 = [(39/4)^2 - 54]/4 = 657/64$ and $p + q = 721$.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2025-I-Q14] | Part C (Q11-Q15)

Answer: 060

1. Cosine rule yields $AC = 7\sqrt{3}$, $AD = 13\sqrt{3}$, with $AC \perp BC$, $AD \perp DE$ and $\angle BAC = \angle DAE = 30^\circ$. For $\theta = \angle CAD$, $\cos\theta = 1/7, \sin\theta = 4\sqrt{3}/7$. Hence $\cos(\theta + 60^\circ) = -11/14$ and $BE = 38$.

2. For triangle ACD, the Fermat-point rotation construction gives $\min_X(AX + CX + DX) = 19\sqrt{3}$: erect an outward equilateral triangle on AC, then the distance from its new vertex to D has square $147 + 507 + 429 = 1083$. All triangle angles are below 120°.

3. Let F be that Fermat point. Its 120° angles put it on the circles ABC and AED, so $\angle AFB = \angle AFE = 90^\circ$. Convexity places B, F, E in that order. Therefore both lower bounds $BX + EX \geq BE$ and $AX + CX + DX \geq 19\sqrt{3}$ are attained at F. The minimum is $38 + 19\sqrt{3}$, giving 60.

Source / 来源

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2025-II-Q15] | Part C (Q11-Q15)

Answer: 240

1. If the minimum is c, then $x(f(x)-c)$ is a monic quartic nonnegative for $x>0$ with two distinct positive double roots, so it equals $(x^2-Sx+P)^2$, with $S,P>0$.

2. Coefficient comparison gives $2S=188+k$, $S^2+2P=10116+188k$, and $P^2=127008k$. Put $k=2t^2$, yielding $S=94+t^2$, $P=504t$. The middle equation becomes $(t-2)(t-4)(t-10)(t+16)=0$.

3. The positive values $t=2,4,10$ give $k=8,32,200$. Each has $S^2>4P$, ensuring two distinct positive roots and making the square factorization sufficient as well. Their sum is 240.

Source / 来源